

Landslide Risk & Hazard Geospatial Data Package

User & Stakeholder Data Sharing Guide | Reference: **NBRI-GEO-PKG-2026**

1. Package Overview

For each target area, the shared package includes four core geospatial components:

- 1. **1:50,000 Landslide Hazard Zonation Map (LHMP_50000)**: Regional-scale hazard zonation polygons depicting relative landslide susceptibility based on geology, Soil thickness and type, slope, hydrology, landuse and landform.
- 2. **Historical Landslide & Instability Incidents (Point Layers)**: Ground-investigated incident records categorized by failure type.
- 3. **Dithawa Cyclone Post-Disaster High-Risk Inspections**: Field inspection points assessed during rapid post-disaster response operations.
- 4. **Mapped Landslide Footprint Polygons**: Delineated spatial boundaries and extent of past landslide events.

2. District Example: Badulla District

Below is an example of the folder structure and dataset contents provided for a single district (**Badulla**):

Directory	Layer / Shapefile Name	Type	Description
01_Hazard_Zonation_Map/	Badulla_LHMP_50000.shp	Polygon	1:50,000 Landslide Hazard Map based on geology, soil thickness & type, slope, hydrology, landuse, and landform.
02_Incident_Inventory_Point s/	Badulla_Cutting_Failure.shp Badulla_Ground_Subsidence.shp Badulla_Landslide.shp Badulla_Rock_Fall.shp Badulla_Slope_Failure.shp Badulla_Other.shp	Point	Cutting Failure (CF), Ground Subsidence (GS), Landslide (L), Rock Fall (RF), Slope Failure (SF), and Other investigated ground instability locations.
03_Dithawa_Cyclone_Inspec tions/	Badulla_Dithawa_HR_Inspections. shp	Point	Cyclone high-risk inspection points assessed during rapid field response operations.
04_Landslide_Footprint_Pol ygons/	Badulla_Polygons.shp	Polygon	Delineated spatial footprints and boundary polygons of historical landslide events.

3. Data Layers & Attribute Dictionaries

A. 1:50,000 Landslide Hazard Zonation Map (LHMP_50000)

Broad-scale planning map dividing the land into relative landslide susceptibility zones based on geology, Soil thickness and type, slope, hydrology, landuse and landform. Format: ESRI Polygon Shapefile | Coordinate System: Sri Lanka Kandawala Grid (EPSG:5234, units: metres).

Attribute Field	Data Type	Description / Interpretation
Range	Numeric / Float	Hazard Rating Category: • 1.0: Landslides not likely to occur / Low Risk • 2.0: Modest level of landslide hazard • 3.0: Landslides are to be expected / High Hazard • 4.0: Very High Hazard / Extreme Risk Areas

B. Historical Incident Inventory (Point Layers)

Field investigations of ground instability conducted across the area, categorized into specific instability types. Format: ESRI Point Shapefiles | Coordinate System: WGS 84 (EPSG:4326, Latitude/Longitude in degrees):

- **Cutting Failure**: Collapse or failure of man-made cut slopes along roads, building sites, or excavations.
- **Ground Subsidence**: Sinking or settling of the ground surface, holes opening up, or ground settlement.
- **Landslide**: Downward movement of a large mass of soil, mud, debris, or rock sliding down a slope or hillside.
- **Rock Fall**: Rocks or boulders breaking loose from steep rock faces or cliffs and rolling or falling down.
- **Slope Failure**: Collapse or failure of natural, un-cut hillside slopes.
- **Other**: Any other ground instability or slope movement reports that do not fit into the five main categories above.

3. Data Layers & Attribute Dictionaries (Continued)

Historical Incident Inventory Attribute Schema (Point Layers):

Field Name	Type	Description	Field Name	Type	Description
Inc_Type	String (50)	Full Incident Name	GND_Name	String (50)	Grama Niladhari Division
Type_Code	String (10)	CF, GS, L, RF, SF, OTHER	Village	String (50)	Village / locality name
Request_No	String (50)	Official NBRI Request ID	Inv_Date	Date / Str	Field investigation date
Case_ID	String (20)	Case tracking ID	Inc_Date	Date / Str	Date incident occurred
District	String (30)	District Name (e.g. Badulla)	Cause	String (254)	Triggering factor / cause
DS_Name	String (50)	Divisional Secretariat	Rain_24h	String (50)	Recorded 24h rainfall (mm)
Latitude	Float	WGS84 Latitude (Y)	Longitude	Float	WGS84 Longitude (X)

C. Dithawa Cyclone High-Risk Inspections (Dithawa_HR_Inspections)

Rapid field inspection locations assessed following the severe Dithawa weather event to identify immediate risk to human life and dwellings. Format: ESRI Point Shapefile | CRS: WGS 84 (EPSG:4326):

Attribute Field	Data Type	Description
Risk_Level	String (20)	Assessed risk level (High Risk, Medium Risk, Low Risk)
HR_Priorit	String (20)	High-Risk Priority classification
Disast_Nat	String (50)	Nature of disaster / hazard identified
Damage_Lvl	String (50)	Observed damage level to building or site
Temp_Recom	String (254)	Temporary field recommendation (evacuation, slope stabilization, monitoring)
Total_Risk	Numeric	Total risk count assessed at the location
Long / Lat	Float	WGS 84 Longitude and Latitude coordinates in Decimal Degrees

D. Mapped Landslide Footprint Polygons (Badulla_Polygons)

Delineated 2D boundary polygons showing the physical footprint, scars, and runout zones of past landslide events. Format: ESRI Polygon Shapefile | CRS: WGS 84 (EPSG:4326):

Attribute Field	Data Type	Description
Name	String (100)	Location or landslide name identifier
Descript	String (254)	Feature description / event notes
District	String (30)	District Name where polygon lies
Layer_Name	String (50)	Mapping survey identification code

4. Coordinate Reference Systems (CRS) & Projection Guide

Dataset	Native Projection	EPSG Code	Units
1:50,000 Hazard Zonation Map	SLD Kandawala Transverse Mercator	EPSG:5234	Metres
Incident Point Layers	WGS 84 (Geographic)	EPSG:4326	Decimal Degrees
Dithawa Cyclone Inspections	WGS 84 (Geographic)	EPSG:4326	Decimal Degrees
Landslide Footprint Polygons	WGS 84 (Geographic)	EPSG:4326	Decimal Degrees

GIS Tip — Automatic Layer Alignment: In modern GIS software (QGIS, ArcGIS Pro, ArcMap), enable **On-the-Fly CRS Transformation** (set project CRS to EPSG:5234 or EPSG:4326). Both datasets will automatically align seamlessly on your map canvas without manual reprojection.

5. Recommended GIS Visualization & Symbology

- **Hazard Zonation Map (LHMP_50000):** Style categorized on Range: Range 1.0 (Pale Green), Range 2.0 (Yellow), Range 3.0 (Orange), Range 4.0 (Red / Crimson). Set opacity to 50–60% over basemaps.
- **Historical Incidents:** Distinct icons by Inc_Type: Landslides (Red dots), Cutting Failures (Blue circles), Rock Falls (Black triangles), Ground Subsidence (Orange squares), Slope Failures (Purple diamonds).
- **Dithawa Cyclone Inspections:** Warning markers on Risk_Level: High Risk (Red), Medium Risk (Yellow), Low Risk (Green).
- **Landslide Polygons:** Semi-transparent red or hatched fill with dark red outline.

6. How to Open in QGIS / ArcGIS Pro

In **QGIS**: Click *Layer* → *Add Layer* → *Add Vector Layer...*, select the .shp files, and set project CRS to **EPSG:5234** or **EPSG:4326**.
In **ArcGIS Pro / ArcMap**: Drag .shp files from Catalog panel into the Map view. On-the-fly coordinate transformation is handled automatically.

For technical inquiries, geospatial queries, or further data requests, please contact the National Building Research Institute (NBRI), Ministry of Defence, Sri Lanka.